Angles on Parallel Lines: Harder

Name _____ Date _____ No calculator Version 1

Scan to practise this topic online

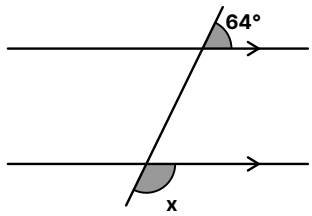
KEY WORDS parallel · transversal · corresponding · alternate · co-interior

A Two steps ●○○

Remember: Sometimes no single rule links the two angles. Find a third angle first, then give both reasons.

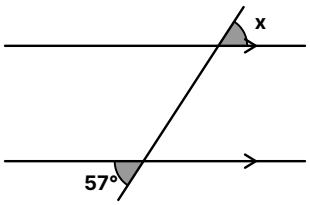
1. The arrows show that the lines are parallel. Find the size of the angle marked x. Give both reasons.

a)



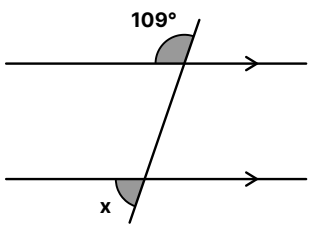
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b)



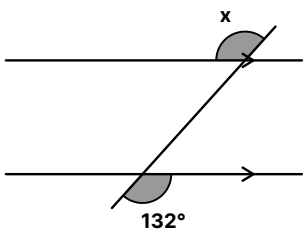
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c)



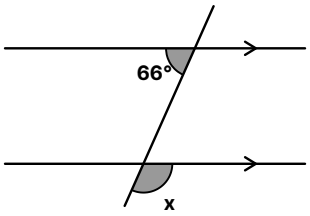
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d)



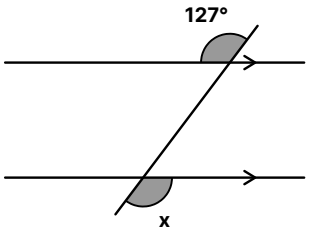
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e)



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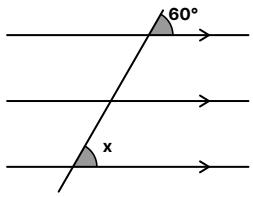
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B Three parallel lines ●●○

Remember: The rules work between any two of the parallel lines, not only neighbours: corresponding angles are equal at every crossing.

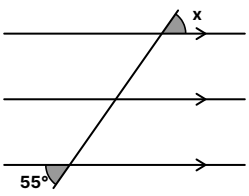
2. All three lines with arrows are parallel. Find the size of the angle marked x.

a)



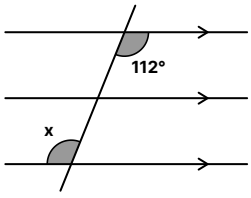
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b)



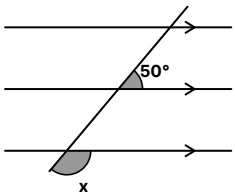
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c)



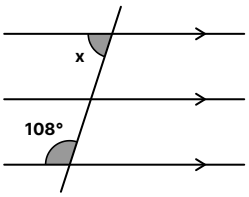
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d)



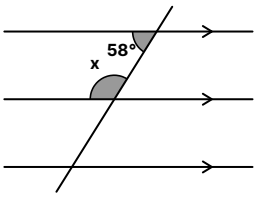
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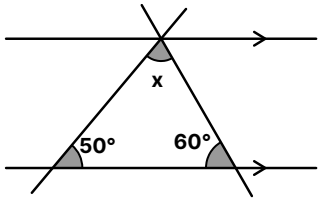
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C A triangle between the lines ●●●

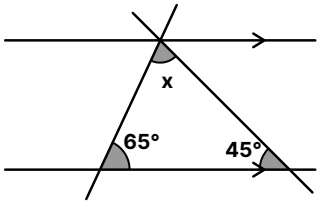
Remember: Two lines crossing the parallels make a triangle. Use alternate angles to move an angle up to the top line, then angles on a straight line, or use the angle sum of a triangle.

3. The two lines with arrows are parallel. Find the angles marked with letters. Give reasons.

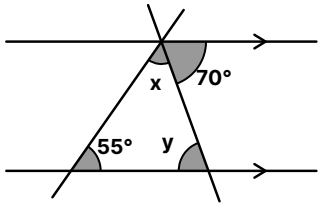
a)



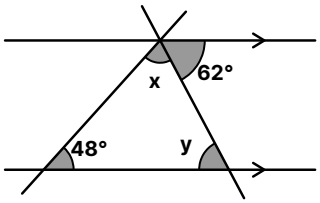
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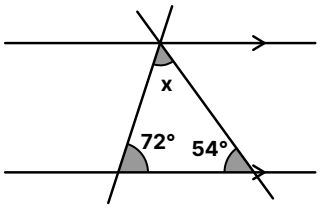
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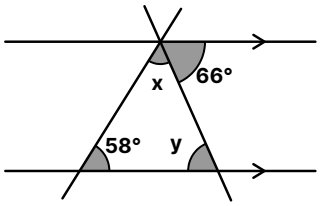
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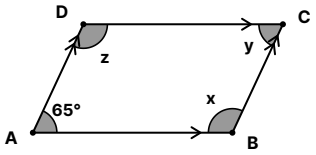


D Parallelograms and trapeziums ●●●

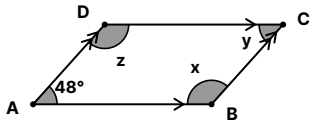
Remember: A parallelogram has two pairs of parallel sides, so angles next to each other are co-interior and add to 180°. A trapezium has one pair.

4. Find the angles marked with letters. Give reasons.

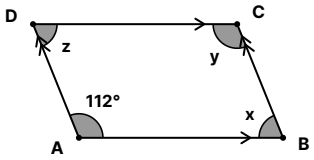
a)



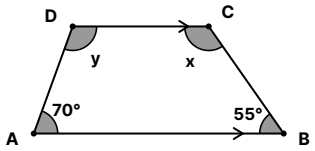
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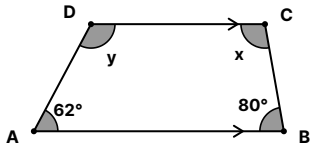
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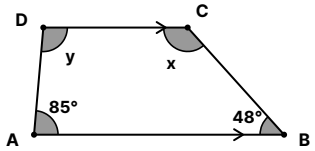
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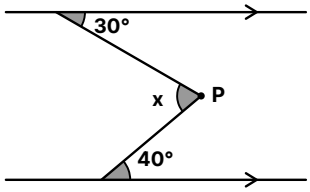


E The zigzag ●●●

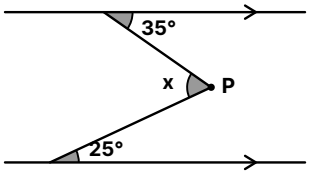
Remember: When a point sits between the parallel lines, draw a third parallel line through it. The angle at the point splits into two alternate angles.

5. The two lines with arrows are parallel. Find the angle marked x.

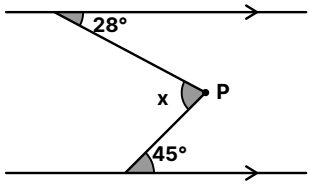
a)



b)

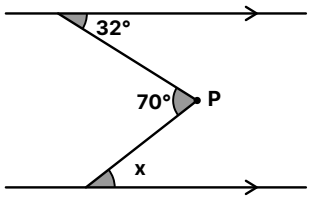


c)

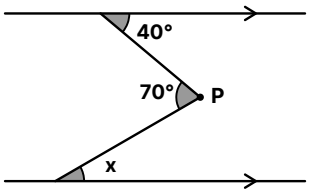


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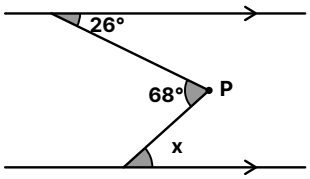
d)



e)



f)

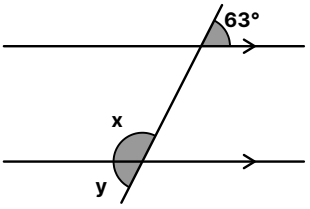


F Two unknowns, with reasons ●●●

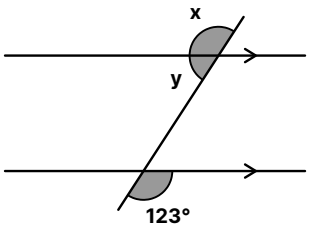
⚠ **Watch out:** Write a reason for each angle, in the order you found them: a reason is only valid if the angle it uses is already known.

6. Find x and y. Give a reason for each.

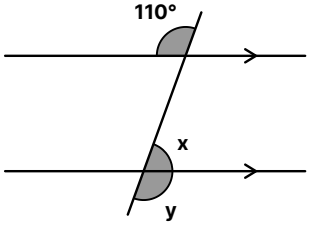
a)



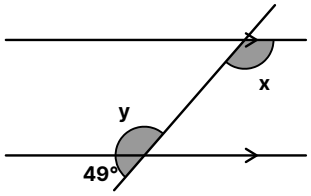
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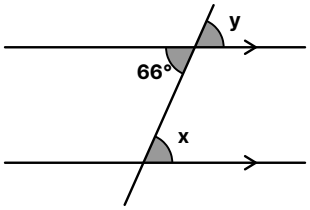
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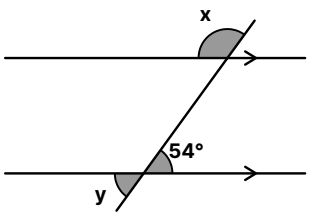
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e)



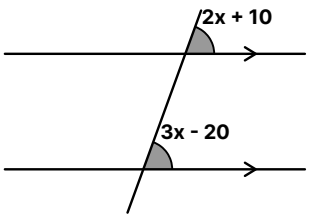
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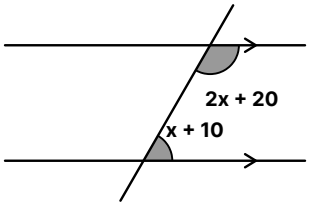
★ Extension

7. Both angles are given in terms of x. Write an equation, solve it, and find the size of each angle.

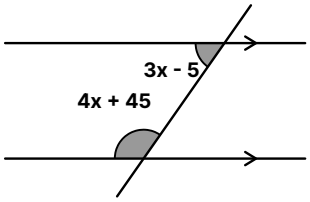
a)



b)



c)



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Angles on Parallel Lines: Harder

Name _____ Date _____ No calculator Version 2

Scan to practise this topic online

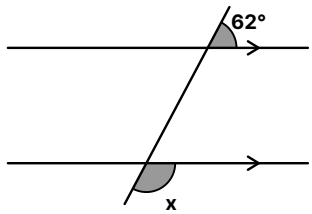
KEY WORDS parallel · transversal · corresponding · alternate · co-interior

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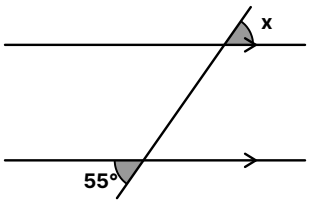
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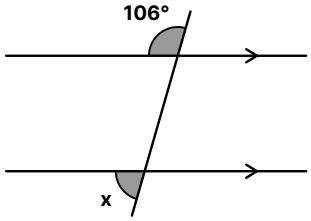
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b)



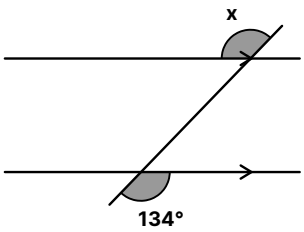
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c)



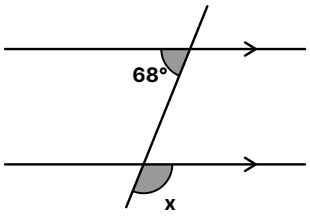
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d)



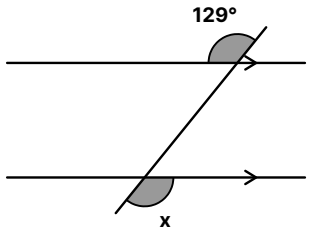
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e)



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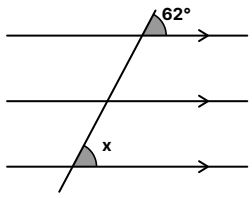
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Remember: The rules work between any two of the parallel lines, not only neighbours: corresponding angles are equal at every crossing.

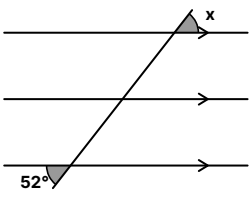
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a)



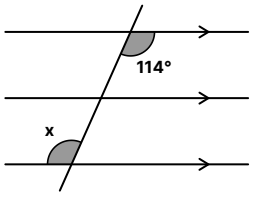
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b)



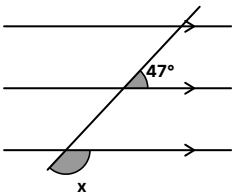
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c)



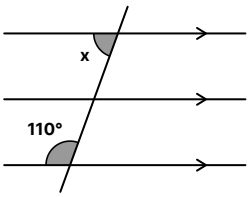
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d)



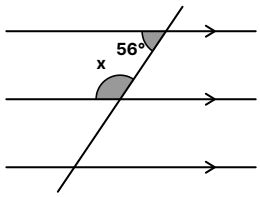
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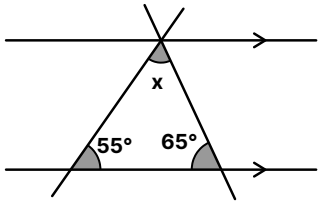
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C A triangle between the lines ●●●

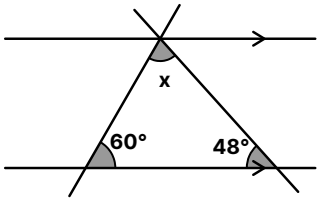
Remember: Two lines crossing the parallels make a triangle. Use alternate angles to move an angle up to the top line, then angles on a straight line, or use the angle sum of a triangle.

3. The two lines with arrows are parallel. Find the angles marked with letters. Give reasons.

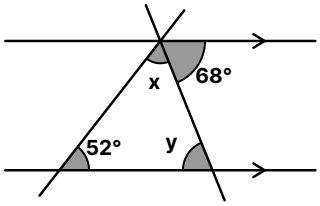
a)



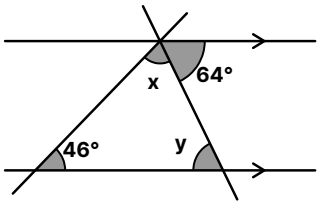
b)



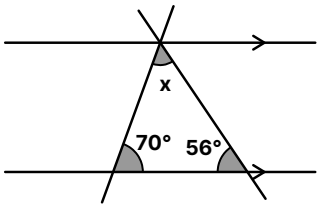
c)



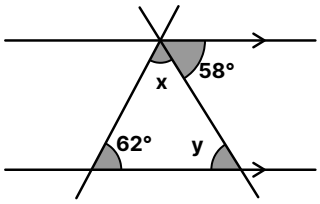
d)



e)



f)

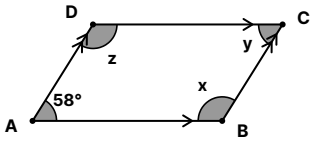


D Parallelograms and trapeziums ●●●

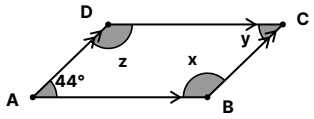
Remember: A parallelogram has two pairs of parallel sides, so angles next to each other are co-interior and add to 180°. A trapezium has one pair.

4. Find the angles marked with letters. Give reasons.

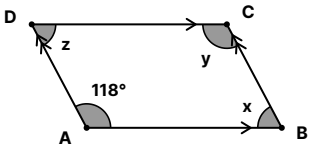
a)



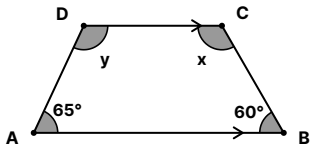
b)



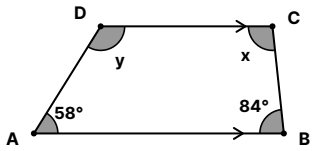
c)



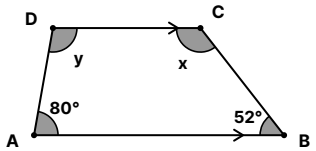
d)



e)



f)

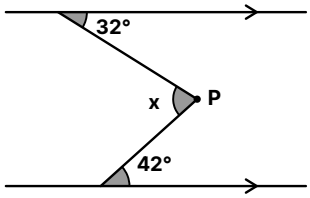


E The zigzag ●●●

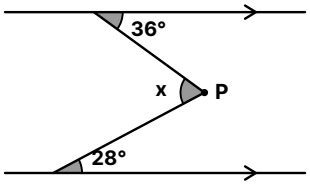
Remember: When a point sits between the parallel lines, draw a third parallel line through it. The angle at the point splits into two alternate angles.

5. The two lines with arrows are parallel. Find the angle marked x.

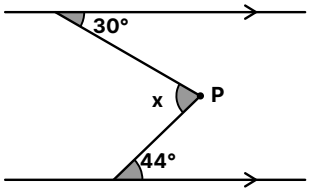
a)



b)

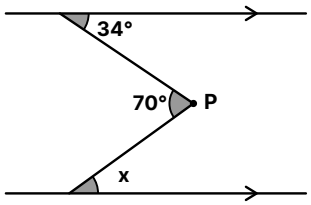


c)

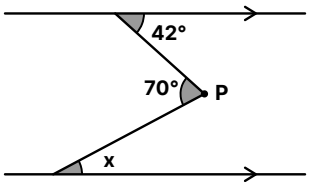


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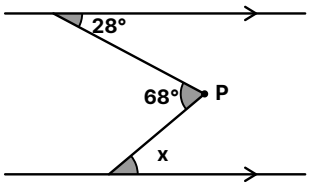
d)



e)



f)

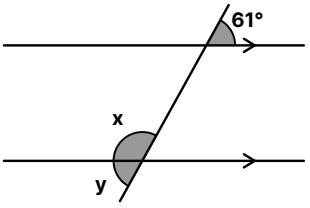


F Two unknowns, with reasons ●●●

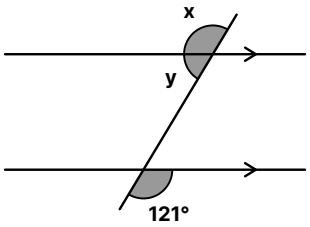
⚠ **Watch out:** Write a reason for each angle, in the order you found them: a reason is only valid if the angle it uses is already known.

6. Find x and y. Give a reason for each.

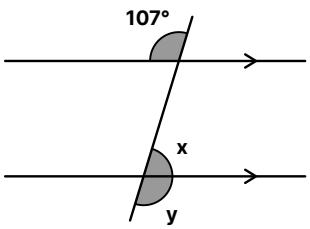
a)



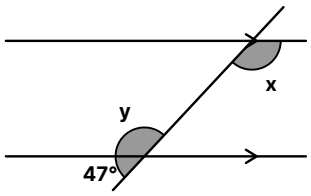
b)



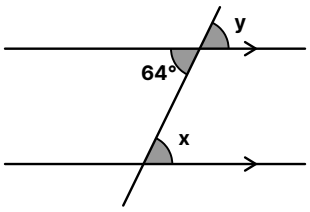
c)



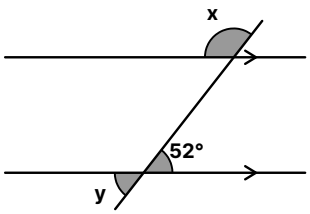
d)



e)



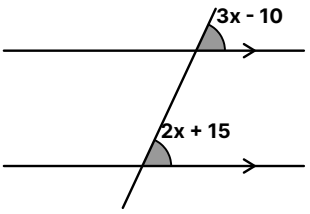
f)



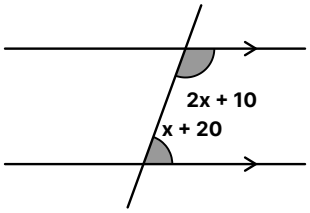
Extension

7. Both angles are given in terms of x. Write an equation, solve it, and find the size of each angle.

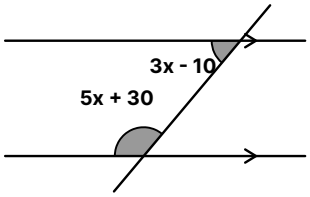
a)



b)



c)



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